

Xinyu (Brian) Guo

GENOMIC FOUNDATION MODELS · CANCER GENOMICS · SCIENTIFIC AI AGENTS

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EDUCATION

Ph.D., Computational Biology & Bioinformatics University of Southern California · 3.95/4.0 · *Viterbi Fellow* 2022 – 2026

M.S., Biostatistics Johns Hopkins University · 3.97/4.0 · *Delta Omega Honor Society* Aug 2020 – May 2022

B.A., Mathematics & Computer Science Washington University in St. Louis · 3.89/4.0 · *Cum Laude* Aug 2018 – May 2020

EXPERIENCE

Postdoctoral Associate — Yale University Sep 2026 – Present

AI for biology & biological foundation models · Advisor: Dr. Lucila Ohno-Machado · New Haven, CT

- Develop and evaluate **biological foundation models** spanning DNA, RNA, and single-cell modalities, focusing on how pretrained representations transfer to variant-effect, regulatory, and disease-risk tasks.

AI Research Scientist Intern — Abbott Cancer Diagnostics (formerly Exact Sciences) Summer 2026

Genomic AI & precision oncology · San Diego (La Jolla), CA

- Researched DNA foundation models, particularly sequence-to-function models such as AlphaGenome and Enigma. Studied these state-of-the-art models against an OncoExTra tumor cohort, **rescuing 2,554 of 6,087 variants (42%)** that conventional snpEff annotation had deprioritized.
- Built **OncoS2F**, a scientific agent that translates natural-language research requests into executable Enigma, Evo2, and Carbon workflows, automating sequence-context construction, batch embeddings, feature extraction, model routing, and provenance across **13 hosted genomic model endpoints**, through to downstream modeling with task-specific classification heads.
- Built FFPE artifact-risk detection models over ~140 GB of paired fresh-frozen and FFPE sequencing data and evaluated them head-to-head against three published tools on a common cohort, reaching **ROC-AUC 0.923** against 0.677 (FFPErase), 0.517 (FFPolish), and 0.509 (FIFA).
- Trained multiple-instance learning models on UNI2-h pathology foundation model embeddings over 2,099 whole-slide images from 1,485 patients, predicting ER (**AUROC 0.890**), TNBC (0.893), and HRD-high (0.790) from H&E under 5-fold × 3-repeat cross-validation, with every leave-institution-out drop ≤ 0.03 ; benchmarked gated-attention against mean-pooling aggregation to settle where the added complexity pays.

Graduate Researcher — University of Southern California Aug 2022 – May 2026

Biological pathology AI/ML modeling · Advisor: Dr. Liang Chen

- Developed a **cross-modality foundation model** integrating DNA sequence, RNA expression, and spatial transcriptomics to identify tissue-specific disease-risk regions, built around explainability so individual genetic variants map to interpretable spatial microenvironments.
- Designed a **self-supervised contrastive learning** method with GNN/GAT architectures for spatial transcriptomics, enabling integration with large-scale multi-omics cohorts across scRNA-seq, scATAC-seq, and spatial modalities for genetic risk evaluation.
- Proposed a fusion architecture using weighted message-passing GNNs and attention to project GWAS/TWAS signals into spatial embeddings, bridging population-scale genetics with tissue-resolved measurement.
- Identified **trans-acting NMD-QTLs across 48 human tissues**, linking common regulatory variation to nonsense-mediated mRNA decay and exposing a mechanism by which non-coding variants modulate transcript stability and disease risk.

Graduate Researcher — Johns Hopkins University Oct 2020 – Aug 2022

Statistical genetics · Advisors: Dr. Nilanjan Chatterjee & Dr. Diptavo Dutta

- Developed **CSTWAS**, a subset-based cross-tissue meta-analysis method for transcriptome-wide association studies that raises power over standard single- and multi-tissue tests while identifying which tissues are actually "active" for a gene-trait signal, improving both power and interpretability.
- Performed transcriptome- and proteome-wide association studies at bulk and single-cell resolution to identify genes linked to **renal cell carcinoma**, and contributed to follow-on work partitioning polygenic risk scores to identify cancer-related proteins.
- Modeled GWAS summary statistics jointly with single-cell ATAC-seq to interpret disease etiology at cellular rather than tissue resolution.

- Built a comprehensive **RNA-seq, ChIP-seq, and ATAC-seq pipeline** supporting differential analysis, gene-set enrichment, and motif discovery, which became shared infrastructure across several lab projects.
- Elucidated how transcriptional and epigenetic regulators mediate hormone-driven tumor resistance to therapeutics, with the chromatin modifier **KMT2D** and its methylation by **SMYD2** as the central axis.

Research Assistant — Washington University in St. Louis

Jun – Aug 2020

- Shipped Lasso and logistic-regression models predicting respiratory failure in COVID-19 patients **into the Barnes-Jewish Hospital EHR**, where they ran against live patient data during the pandemic peak to flag high-risk patients for ICU triage under acute bed and ventilator constraints.

SELECTED PROJECTS

- Benchmarked **AlphaGenome** against atrial-fibrillation TWAS 2025 · xinyuguo.com/projects/ag-twas-alphagenome — scored sequence-model predictions on 858 gene–tissue pairs and 10,120 fine-mapped variant–gene–tissue triplets across 10 GTEx tissues; chromatin tracks enriched for causal variants (ATAC 1.18×, $P < 10^{-4}$) while expression predictions held only 55–60% sign accuracy, mapping where sequence models can and cannot substitute for genotype-trained methods.
- Trained a **560M-parameter decoder-only LLM from scratch** 2025 · xinyuguo.com/projects/nanochat-llm-training — built the full pipeline end to end: Rust BPE tokenizer, data loading, base **pretraining**, midtraining, **supervised fine-tuning**, and **reinforcement learning**, with distributed PyTorch DDP across 8 GPUs, mixed precision, and gradient-synchronization tuning for cluster throughput; evaluated on CORE, ARC, MMLU, GSM8K, and HumanEval.
- State-space sequence models for long-horizon and edge deployment 2025 – 2026 · lead developer / principal architect — **TAL-Mamba**, temporal action localization over 2+ hour sequences with efficient state management, cutting memory 68% versus Transformer baselines; and **Track Mamba**, a linear-complexity real-time small-object tracker 40% smaller and 50% faster than TrackNet baselines at higher precision–recall.
- **Legal Patent Copilot** 2024 — retrieval agent for prior-art discovery over 10M+ documents using FAISS indexing and fine-tuned reranking layers, cutting search latency **42%** with improved recall.

SELECTED PUBLICATIONS

1. Guo, X., Chen, L. From G1 to M: a comparative study of methods for identifying cell cycle phases. *Briefings in Bioinformatics* (2024).
2. Guo, X., Chatterjee, N., Dutta, D. Subset-based method for cross-tissue transcriptome-wide association studies improves power and interpretability. *Human Genetics and Genomics Advances* (2024).
3. Dutta, D., Guo, X., Winter, T.D., et al. Transcriptome- and proteome-wide association studies identify genes associated with renal cell carcinoma. *The American Journal of Human Genetics* (2024).
4. Bianchi, A., De Castro Silva, I., ... Guo, X., et al. Cell-autonomous Cxcl1 sustains tolerogenic circuitries and stromal inflammation via neutrophil-derived TNF in pancreatic cancer. *Cancer Discovery* (2023).
5. Blawski, R., Vokshi, B.H., Guo, X., et al. Methylation of the chromatin modifier KMT2D by SMYD2 contributes to therapeutic response in hormone-dependent breast cancer. *Cell Reports* (2024).
6. Kittane, S., Ladewig, E., ... Guo, X., et al. The histone methyltransferase KMT2D mediates subtype-specific transcriptional regulation, lineage plasticity, and therapeutic response in castration-resistant prostate cancer. *Cancer Research* (2024, 2025).
7. Jagirdar, K., Portuallo, M.E., ... Guo, X., et al. ERK hyperactivation as a unified mechanism of escape in intrinsic and acquired CDK4/6-inhibitor resistance in acral lentiginous melanoma. *Oncogene* (2024).
8. Dutta, D., Zhang, J., Guo, X., et al. Leveraging and partitioning polygenic risk scores to identify cancer-related proteins. *medRxiv* (2025, preprint).

TECHNICAL EXPERTISE

AI / ML Genomic foundation models (AlphaGenome, Enigma, Evo2, Carbon, Borzoi, Enformer, DNABERT) · transformers · state-space models (Mamba) · GNN/GAT · self-supervised & contrastive learning · LLM pretraining · agentic systems & tool use · RAG / vector retrieval · benchmark design & leakage-safe evaluation

COMP. BIOLOGY Cancer genomics & precision oncology · variant-effect prediction (SNV, SV, fusion) · regulatory genomics & splicing · GWAS/TWAS · single-cell and spatial transcriptomics · RNA-seq / ChIP-seq / ATAC-seq · NGS somatic calling, FFPE artifacts, VCF · Ensembl, NCBI, UniProt, PDB, OncoKB, CIViC

ENGINEERING Python · R · C++ · SQL · Bash · TypeScript · PyTorch (DDP/FSDP), PyG, Hugging Face, JAX, scikit-learn, CUDA · FastAPI · React · Nextflow / WDL · DuckDB · Docker, Git

CLOUD / SCALE AWS (SageMaker, Bedrock, HealthOmics, EC2/ECS/S3) · petabyte-scale data on HPC · multi-GPU training